A controlled IVF/SPEA2 hybridization under fixed evaluation budgets

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Abstract

Multi-objective optimization problems (MOPs) require approximating a set of non-dominated solutions that balances conflicting objectives under limited evaluation budgets. Although SPEA2 remains a widely adopted baseline due to its elitist archiving and density-aware fitness assignment, its search dynamics can stagnate on challenging landscapes where locally promising regions become penalized by density pressure. This paper investigates a controlled hybridization of SPEA2 with the In Vitro Fertilization (IVF) procedure, yielding IVF/SPEA2, in which IVF is activated under a fixed function evaluation budget and aligned with SPEA2's fitness-and-archiving logic through a relaxed acceptance criterion based on SPEA2 fitness ranking. Experiments were conducted in PlatEMO using ZDT, DTLZ, WFG, and MaF benchmark suites. Performance was assessed using IGD and validated via the Wilcoxon rank-sum test over 100 independent runs per configuration. The results indicate that IVF/SPEA2 achieves statistically significant improvements over canonical SPEA2 in a substantial fraction of instances while maintaining competitive behavior across diverse problem classes, with gains remaining problem-dependent. These findings support that a carefully budgeted memetic intensification can complement SPEA2 without disrupting its diversity mechanisms, and motivate further investigation on parameter adaptivity and operator-level sensitivity.

Keywords: memetic computing, SPEA2, in vitro fertilization, multiobjective optimization, Pareto front

1 Introduction

Multiobjective Optimization Problems (MOPs) arise in a wide range of scientific and engineering domains, including design optimization, logistics, network planning, and data-driven decision making. Such problems are characterized by the presence of conflicting objectives, for which no single optimal solution exists. Instead, the goal is to approximate the Pareto front, composed of non-dominated solutions representing different trade-offs among objectives. Evolutionary Multiobjective Algorithms (MOEAs) have become a dominant class of methods for tackling MOPs due to their population-based nature and their ability to approximate diverse regions of the Pareto front in a single run Deb et al. (2002); Zhou et al. (2011).

Among classical MOEAs, algorithms such as NSGA-II Deb et al. (2002), MOEA/D Zhang and Li (2007), and SPEA2 Zitzler et al. (2001) have played a central role in shaping the field. In particular, SPEA2 combines elitist archiving, fitness assignment based on dominance relations, and an explicit density estimation mechanism to preserve diversity Zitzler et al. (2001). Despite its age, SPEA2 remains relevant in contemporary applications and continues to be adopted as a baseline or reference method in both academic studies and practical scenarios, including recent applied optimization problems Gadhvi et al. (2016). This sustained relevance motivates ongoing research into principled extensions and variants that seek to improve its performance while preserving its conceptual foundations.

However, SPEA2 is known to exhibit limitations in certain classes of problems, particularly those characterized by multimodality, deceptive landscapes, or complex geometries of the Pareto front. In such settings, the interaction between dominance-based fitness assignment and density estimation may lead to slow convergence or premature stagnation, as locally promising regions can be penalized due to high solution density. These issues have motivated several extensions and hybridizations of SPEA2, including alternative density estimation strategies and auxiliary mechanisms designed to improve the balance between convergence and diversity Bosman and Thierens (2003); Li et al. (2013).

Memetic algorithms and hybrid evolutionary approaches provide a natural framework for addressing such limitations. By integrating evolutionary search with additional intensification mechanisms, memetic algorithms aim to enhance search efficiency while retaining population diversity. Within this context, the In Vitro Fertilization (IVF) procedure was proposed as a biologically inspired population-based intensification method Camilo-Junior and Yamanaka (2011). Subsequent studies demonstrated that coupling IVF with different MOEAs can improve convergence and diversity on benchmark problems, including IVF/NSGA-II Sampaio and Camilo (2017) and more recent extensions such as IVF/NSGA-III Sampaio et al. (2023). These results suggest that IVF constitutes a flexible and effective memetic component when appropriately integrated into a host algorithm.

Despite these advances, the interaction between IVF and the internal dynamics of different MOEAs is not uniform and cannot be assumed to be universally beneficial. In particular, directly coupling IVF to SPEA2 raises nontrivial design challenges, as the local intensification induced by IVF may conflict with the density-based fitness assignment and archive truncation mechanisms of SPEA2. Without careful control,

offspring generated by IVF may be systematically penalized due to increased local density, potentially negating the intended benefits of intensification.

This work investigates a controlled hybridization of SPEA2 with an IVF module, referred to as IVF/SPEA2, with the objective of understanding when and why such a coupling can be beneficial. Rather than proposing an ad hoc integration, the hybridization is treated as a principled design decision, in which the frequency of IVF activation, the size of the IVF collection, and the criterion used to select candidate offspring are explicitly aligned with the fitness assignment and archiving logic of SPEA2. In particular, a relaxed selection rule is adopted to mitigate density-induced penalization while maintaining compatibility with the host algorithm.

The central hypothesis of this study is that the controlled intensification provided by IVF can complement the global search capabilities of SPEA2 under a fixed evaluation budget, leading to improved Pareto front approximation in specific classes of problems. At the same time, it is explicitly assumed that such improvements are problem-dependent and sensitive to key IVF parameters, and therefore cannot be generalized across all MOPs. Accordingly, this work focuses on benchmark-based evaluation and emphasizes mechanistic interpretation of both successful and unsuccessful cases.

The contributions of this paper are threefold. First, it presents a principled IVF/SPEA2 hybridization strategy that explicitly accounts for the interaction between IVF-based intensification and SPEA2's density-based fitness assignment. Second, it provides a systematic experimental evaluation on widely used benchmark suites, highlighting both gains and failures relative to canonical SPEA2 under identical evaluation budgets. Third, it frames parameter sensitivity and problem dependence as intrinsic properties of the hybrid method, thereby delineating its scope of validity and positioning it clearly within the landscape of memetic and hybrid evolutionary algorithms.

2 Related Work

2.1 SPEA2 and density-aware archiving

SPEA2 is a canonical multi-objective evolutionary algorithm that combines elitist selection with an external archive and a fitness assignment that integrates dominance strength and density estimation. Its environmental selection and truncation mechanisms aim to preserve a diverse approximation of the Pareto front under a fixed population and archive size. Despite its continued relevance as a reference method, SPEA2 can exhibit slow convergence or stagnation on landscapes where density pressure and truncation interact unfavorably with locally intensified search, especially when the Pareto set geometry induces clustering or deceptive attraction basins.

2.2 Memetic algorithms in multi-objective optimization

Memetic algorithms hybridize population-based evolutionary search with additional intensification components, typically to accelerate convergence while maintaining diversity. In the multi-objective setting, memetic mechanisms are frequently coupled

to established MOEAs to improve solution quality under limited budgets. However, the effectiveness of a memetic component is not uniform across host algorithms, because intensification can conflict with the host's diversity-preservation and selection logic. Therefore, principled couplings that respect the host algorithm's internal fitness dynamics are essential to avoid systematic penalization of intensified offspring.

2.3 The IVF procedure and IVF-based hybridizations

The In Vitro Fertilization (IVF) procedure was originally introduced as an auxiliary intensification method that operates on a selected collection of individuals through specialized recombination operators and controlled activation. Prior work reported IVF hybridizations with different MOEAs, such as IVF/NSGA-II and IVF/GDE3, indicating that IVF can improve convergence and diversity on benchmark problems when appropriately integrated. IVF is typically controlled by parameters such as an execution rate and a collection size, which regulate how often the procedure is invoked and how many donors are involved.

2.4 Motivation for a controlled IVF/SPEA2 coupling

Directly coupling IVF to SPEA2 raises a specific challenge: SPEA2's density-aware fitness and archive truncation can systematically penalize locally intensified solutions, potentially neutralizing the benefits of IVF. This motivates a controlled coupling in which IVF activation is budgeted and the acceptance of IVF-derived offspring is aligned with SPEA2's fitness-and-archiving logic. The present work addresses this gap by proposing an IVF/SPEA2 coupling that preserves the canonical SPEA2 flow while introducing a relaxed fitness-based acceptance criterion to mitigate density-induced penalization.

3 Proposed IVF/SPEA2 Method

This section presents IVF/SPEA2, a hybrid algorithm that integrates the In Vitro Fertilization (IVF) procedure into SPEA2 to enhance convergence and diversity in multi-objective optimization. The hybridization preserves the canonical SPEA2 operational flow while introducing a controlled coupling mechanism at the point of new solution generation. Fig. 1 illustrates the complete operational flow; the key design element is the conditional IVF activation at step (4), which determines whether the current generation includes IVF-derived offspring or relies exclusively on canonical SPEA2 variation.

The IVF parameters, namely the Execution Rate r and the Collection Size c, govern the balance between generating new individuals and the intensity of the auxiliary genetic manipulations. Prior studies and empirical evidence suggest default values around 10% for both parameters, which provides a practical balance between improvement potential and function evaluation consumption. In IVF/SPEA2, these parameters allow controlled modulation of IVF interference, ensuring the module supports (rather than disrupts) the host evolutionary dynamics.

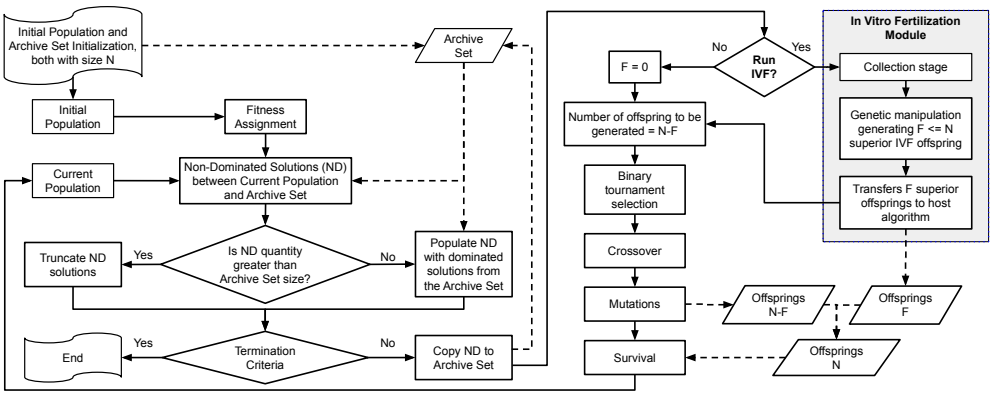

Fig. 1 Operational flow of the IVF/SPEA2 algorithm. Steps (1)–(3) and (5)–(6) follow canonical SPEA2; step (4) introduces the conditional IVF activation under a fixed evaluation budget.

3.1 IVF operator used in this study

The IVF procedure admits different genetic operators, such as Assisted Recombination (AR) and exploratory variants (EAR-N, EAR-T, EAR-P, EAR-PA). To avoid confounding effects and to preserve a controlled interpretation of the coupling with SPEA2 under a fixed evaluation budget, the experiments reported in this paper instantiate IVF using **Assisted Recombination (AR)** as the IVF operator. Exploratory variants are left for future operator-level analyses.

3.2 Coupling Strategy and Function Evaluation Budget

The coupling mechanism is integrated just before the host algorithm generates its traditional offspring. In each generation, IVF/SPEA2 maintains a fixed population size N by allocating a portion of the generation's evaluation budget to the IVF module. Let F denote the number of offspring produced by IVF during a generation, constrained by $F \leq N$. The host algorithm then generates $N - F$ additional offspring using the standard SPEA2 reproduction operators, ensuring that exactly N offspring are produced per generation in total.

The activation of IVF in a given generation is governed by the total Function Evaluations (FEs) consumed so far and by the execution rate parameter r, which represents the percentage of total evaluations that IVF will consume. This mechanism allows some generations to remain unmodified, reproducing the behavior of canonical SPEA2, while enabling calibrated intervention when activated.

3.3 Adaptations for SPEA2 Integration

A central design challenge in coupling IVF to SPEA2 arises from SPEA2's density-aware selection and archive truncation mechanisms. Unlike dominance-and-crowding approaches (e.g., NSGA-II), SPEA2 assigns fitness using both dominance strength and density estimation. Therefore, the numerically best fitness value does not necessarily represent the most suitable individual under density pressure.

Algorithm 1 IVF/SPEA2 under a fixed function evaluation budget

Require: Population size N, archive size $\bar{N}$ ($\bar{N} = N$), max evaluations $FE_{\max}$, execution rate r, collection size c, IVF operator (AR)

Initialize population P_0 with N solutions; evaluate; $FE \leftarrow N$
Initialize archive $\bar{P}_0 \leftarrow \emptyset$
while $FE < FE_{\max}$ **do**
  Compute SPEA2 fitness on $P_t \cup \bar{P}_t$; update archive $\bar{P}_{t+1}$ via environmental selection and truncation
  $F \leftarrow 0$; $O \leftarrow \emptyset$
  if IVF-TRIGGER$(FE, FE_{\max}, r)$ is true **then**
    Select a donor collection C of size $\lceil c \cdot N \rceil$ from $P_t \cup \bar{P}_{t+1}$ using SPEA2 fitness ranking
    while $F < N$ **and** IVF budget for current generation not exceeded **do**
      Generate candidate offspring x from C using IVF operator AR; evaluate x; $FE \leftarrow FE + 1$
      if x ranks among the best $2\lceil c \cdot N \rceil$ solutions of $P_t \cup \bar{P}_{t+1} \cup \{x\}$ by SPEA2 fitness **then**
        $O \leftarrow O \cup \{x\}$; $F \leftarrow F + 1$
      end if
    end while
  end if
  Generate $N - F$ offspring using standard SPEA2 variation; evaluate; $FE \leftarrow FE + (N - F)$
  Set P_{t+1} from the evaluated offspring set of size N
  $t \leftarrow t + 1$
end while
return Final archive $\bar{P}_t$

To better align IVF cycles with SPEA2's internal logic, IVF/SPEA2 modifies the comparison strategy used to decide whether a newly derived individual should trigger a new IVF cycle. Instead of a direct Pareto dominance comparison, the method verifies whether the derived solution is among the $2 \times c$ best individuals (i.e., those with the lowest fitness values) in the complete known set of solutions. The complete set contains $N + \bar{N}$ individuals, where N is the population size and $\bar{N}$ is the archive size (with $N = \bar{N}$ in SPEA2).

The selection criterion relies on SPEA2's fitness:

$$F(i) = R(i) + D(i),$$

where

$$R(i) = \sum_{j \in P_t \cup \bar{P}_t,\ j \prec i} S(j)$$

is the raw fitness based on dominance strength, and

$$D(i) = \frac{1}{\sigma_i^k + 2}$$

is the density estimation based on the k-th nearest neighbor distance. The fitness verification checks whether the generated solution belongs to the $2 \times c$ best solutions according to $F(i)$. If the condition is met, a new IVF cycle is initiated, using the derived solution with the best (lowest) $F(i)$ as the primary donor for subsequent operations. This adaptation maintains consistency with SPEA2's diversity-and-convergence philosophy while mitigating density-induced penalization of locally intensified solutions.

3.4 IVF Module Stages

The IVF method operates through four stages within IVF/SPEA2:

3.4.1 Activation Trigger

The activation trigger determines when IVF executes during evolution. Controlled by r, it activates according to the number of Function Evaluations (FE) already consumed. This preserves canonical SPEA2 behavior for non-activated generations and provides fine-grained control of IVF intervention frequency.

3.4.2 Collection Stage

The collection stage selects high-quality genetic material from the current population according to SPEA2 fitness. The collection size is controlled by c, where larger values provide more material and potentially higher diversity, but also increase evaluation consumption.

3.4.3 Genetic Manipulation

The genetic manipulation stage applies specialized genetic operations on the collected solutions, enabling both intensification (subtle modifications of known solutions) and diversification (generation of novel solutions). The module can perform multiple cycles while respecting the constraint $F \leq N$ in each generation.

3.4.4 Transfer Stage

The transfer stage integrates superior IVF-derived solutions into the host algorithm. A solution is eligible for transfer if it ranks among the $2 \times c$ best individuals of the complete known set according to SPEA2 fitness $F(i)$. The module transfers F superior offspring and the host generates $N - F$ standard offspring, maintaining a constant population size per generation.

Overall, IVF/SPEA2 preserves SPEA2's elitist evolutionary flow while introducing a controlled memetic component at the reproduction stage, explicitly aligned with SPEA2's density-aware fitness assignment.

4 Experimental Setup

4.1 Platform and Algorithms

Experiments were conducted using the PlatEMO platform Tian et al. (2017) in MATLAB to compare the proposed IVF/SPEA2 algorithm with established multi-objective evolutionary algorithms, namely SPEA2 Zitzler et al. (2001), MFO-SPEA2, SPEA2+SDE, NSGA-II Deb et al. (2002), NSGA-III Deb and Jain (2014), and MOEA/D Zhang and Li (2007). The benchmark algorithms were executed using their default parameter settings as defined in PlatEMO version 24.2.0.2923080.

For the IVF module in IVF/SPEA2, default configurations grounded in prior studies and experimental findings were adopted: Collection Size $c = 10\%$ of the population size and Execution Rate $r = 10\%$.

For IVF/SPEA2, the IVF operator instantiated in the experiments is Assisted Recombination (AR), with default IVF parameters $c = 10\%$ and $r = 10\%$.

4.2 Benchmark Suites

The evaluation encompassed multiple benchmark suites selected for complementary characteristics and varying complexity levels:

- **ZDT** offers bi-objective problems with diverse Pareto front geometries (convex, concave, disconnected).
- **DTLZ** extends evaluation to scalable objective counts and increasing dimensionality.
- **WFG** introduces transformations yielding complex geometries with characteristics such as non-separability, multimodality, and asymmetry.
- **MaF** provides many-objective optimization problems that challenge traditional approaches.

4.3 Performance Metric

Performance was assessed using the Inverted Generational Distance (IGD), a widely adopted indicator measuring both convergence and diversity by computing the distance between the obtained Pareto front approximation and a reference set representing the true Pareto-optimal front. Lower IGD values indicate better performance, reflecting solutions that are both well-distributed and close to the optimal front.

4.4 Statistical Analysis and Protocol

For statistical validation, the Wilcoxon rank-sum test was applied to compare algorithmic performance using IGD values. Statistical significance was established at $p < 0.05$.

Due to the stochastic nature of the algorithms, each test configuration was executed 100 times for IVF/SPEA2 and for each compared algorithm (SPEA2, MFO-SPEA2, SPEA2+SDE, NSGA-II, NSGA-III, and MOEA/D), with performance data collected per run using IGD.

Table 1 Summary of results: IVF/SPEA2 vs. SPEA2.

M	Instances	Wins	Losses	Win proportion (%)
2	28	16	0	57.1
3	23	14	3	60.9

4.5 Threats to validity

Internal validity. IVF/SPEA2 adopts default IVF parameters ($c = 10\%$, $r = 10\%$) grounded in prior IVF studies. Although this choice supports a controlled evaluation under a fixed budget, different parameterizations could alter the observed gains. To mitigate stochastic effects, all algorithms were executed 100 independent runs per configuration and compared through nonparametric statistical testing.

Construct validity. Performance is measured using IGD, which captures both convergence and diversity relative to a reference set. While IGD is widely used in multi-objective optimization, conclusions may vary under alternative indicators (e.g., hypervolume) or under different reference set resolutions.

Conclusion validity. The Wilcoxon rank-sum test with $p < 0.05$ is used to assess significance across runs. Multiple comparisons across many instances can inflate false discovery rates; therefore, results are interpreted primarily as evidence of problem-dependent behavior rather than universal dominance.

External validity. The evaluation focuses on synthetic benchmark suites (ZDT, DTLZ, WFG, MaF), which provide controlled diversity of front geometries but may not fully represent real-world objective interactions. Additionally, the IVF/SPEA2 implementation within PlatEMO improves reproducibility but framework-level choices (including IVF operator details) can influence outcomes; using a standard platform mitigates this risk.

5 Results

This section reports the empirical performance of IVF/SPEA2 under the default IVF settings ($c = 10\%$, $r = 10\%$), first against its canonical host (SPEA2) and then in comparison to other established algorithms.

5.1 Primary Comparison: IVF/SPEA2 vs. SPEA2

To establish fundamental performance characteristics, we first present a direct comparison between IVF/SPEA2 and SPEA2. The win criterion is defined as a statistically significant improvement according to the Wilcoxon rank-sum test ($p < 0.05$) applied to IGD values.

Across bi-objective instances ($M = 2$), IVF/SPEA2 achieves statistically significant improvements in 16 out of 28 test instances (57.1%) with no statistically significant degradations. For tri-objective instances ($M = 3$), IVF/SPEA2 obtains 14 wins (60.9%) against 3 losses (13.0%). These results indicate that IVF/SPEA2 can improve SPEA2 in many cases, while its benefits remain problem-dependent.

Table 2 Comparison of algorithms vs. SPEA2 using IGD and Wilcoxon rank-sum test ($p < 0.05$).

Algorithm	+	−	=
$M = 2$			
IVF/SPEA2	16	0	12
MFO-SPEA2	6	0	22
SPEA2+SDE	7	0	21
NSGA-II	13	11	4
NSGA-III	18	9	1
MOEA/D	17	11	0
$M = 3$			
IVF/SPEA2	14	3	6
MFO-SPEA2	6	2	15
SPEA2+SDE	6	2	15
NSGA-II	9	11	3
NSGA-III	7	16	0
MOEA/D	10	13	0

5.2 Comparison Against Established Algorithms

To contextualize performance within the broader landscape of MOEAs, we compare IVF/SPEA2 against NSGA-II, NSGA-III, MOEA/D, MFO-SPEA2, and SPEA2+SDE, using SPEA2 as the reference baseline. The comparison matrix summarizes wins (+), losses (-), and ties (=) based on the Wilcoxon rank-sum test applied to IGD.

The comparative analysis shows that IVF/SPEA2 exhibits the most consistent positive performance against the SPEA2 baseline across both two- and three-objective cases. While NSGA-III performs strongly in the bi-objective setting, its performance declines in the tri-objective scenario. In contrast, IVF/SPEA2 maintains robust performance in both cases, indicating favorable behavior across the evaluated dimensionalities.

5.3 Detailed Results Across Benchmark Suites

Detailed results for bi-objective and tri-objective settings are reported as median IGD values per benchmark instance, including the experimental configuration fields (Problem, N, M, D, and FE). Statistical indicators adjacent to IVF/SPEA2 values denote statistically significant superiority (+), equivalence (=), or inferiority (-) relative to SPEA2 under the Wilcoxon rank-sum test ($p < 0.05$).

In the bi-objective setting, IVF/SPEA2 achieves statistically significant improvements in 16 out of 28 instances and shows no statistically significant deterioration, demonstrating robustness across diverse landscapes. The method matches or exceeds SPEA2 on classical ZDT and DTLZ benchmarks and remains competitive against the reference algorithms.

Table 3 Comparison of IGD (median) between IVF/SPEA2 and baselines for $M = 2$ (ZDT1–ZDT6, DTLZ1–DTLZ7, WFG1–WFG9, and MaF1–MaF7). Symbols +, −, and = denote statistically significant superiority, inferiority, or equivalence relative to SPEA2 under the Wilcoxon rank-sum test ($p < 0.05$).

Problem	N	M	D	FE	IVF/SPEA2	MFO-SPEA2	SPEA2+SDE	NSGA-II	NSG
ZDT1	100	2	30	100000	3.9091×10^{-3} '+'	3.9324×10^{-3} '+'	4.2415×10^{-3} '−'	4.8135×10^{-3} '−'	$3.8879 \times 10^{-}$
ZDT2	100	2	30	100000	3.9041×10^{-3} '='	3.9269×10^{-3} '='	6.0190×10^{-3} '−'	4.8843×10^{-3} '−'	$3.8070 \times 10^{-}$
ZDT3	100	2	30	100000	4.8535×10^{-3} '+'	4.8859×10^{-3} '='	5.2992×10^{-3} '−'	5.4302×10^{-3} '−'	$6.1267 \times 10^{-}$
ZDT4	100	2	10	100000	3.8438×10^{-3} '+'	3.8335×10^{-3} '+'	4.2477×10^{-3} '−'	4.5402×10^{-3} '−'	$3.8949 \times 10^{-}$
ZDT6	100	2	10	100000	3.0782×10^{-3} '+'	3.0847×10^{-3} '+'	3.4256×10^{-3} '−'	3.6734×10^{-3} '−'	$3.0013 \times 10^{-}$
DTLZ1	100	2	6	100000	1.8274×10^{-3} '+'	1.8366×10^{-3} '='	1.9998×10^{-3} '−'	2.2062×10^{-3} '−'	$1.7860 \times 10^{-}$
DTLZ2	100	2	11	100000	4.1387×10^{-3} '+'	4.1742×10^{-3} '='	9.6713×10^{-3} '−'	5.1248×10^{-3} '−'	$3.9660 \times 10^{-}$
DTLZ3	100	2	11	100000	4.1941×10^{-3} '+'	4.2814×10^{-3} '='	1.0069×10^{-2} '−'	5.3101×10^{-3} '−'	$4.2241 \times 10^{-}$
DTLZ4	100	2	11	100000	4.1576×10^{-3} '='	4.1664×10^{-3} '='	9.8278×10^{-3} '−'	5.2006×10^{-3} '−'	$3.9662 \times 10^{-}$
DTLZ5	100	2	11	100000	4.1432×10^{-3} '+'	4.1708×10^{-3} '='	1.0040×10^{-2} '−'	5.0335×10^{-3} '−'	$3.9660 \times 10^{-}$
DTLZ6	100	2	11	100000	4.0691×10^{-3} '+'	4.0862×10^{-3} '='	9.5098×10^{-3} '−'	5.6503×10^{-3} '−'	$3.9660 \times 10^{-}$
DTLZ7	100	2	21	100000	4.7817×10^{-3} '='	4.7290×10^{-3} '='	5.2461×10^{-3} '−'	5.3784×10^{-3} '−'	$5.1155 \times 10^{-}$
WFG1	100	2	11	100000	1.2123×10^{-2} '+'	1.0933×10^{-1} '−'	4.0805×10^{-2} '−'	2.1513×10^{-2} '+'	$9.6925 \times 10^{-}$
WFG2	100	2	11	100000	1.1092×10^{-2} '='	1.0983×10^{-2} '='	1.3551×10^{-2} '−'	1.2888×10^{-2} '−'	$1.3849 \times 10^{-}$
WFG3	100	2	11	100000	1.1996×10^{-2} '='	1.1975×10^{-2} '='	1.3096×10^{-2} '−'	1.4471×10^{-2} '−'	$1.1343 \times 10^{-}$
WFG4	100	2	11	100000	1.2761×10^{-2} '+'	1.2921×10^{-2} '='	3.2335×10^{-2} '−'	1.5283×10^{-2} '−'	$1.2233 \times 10^{-}$
WFG5	100	2	11	100000	6.3535×10^{-2} '+'	6.3653×10^{-2} '='	7.5654×10^{-2} '−'	6.5339×10^{-2} '−'	$6.3370 \times 10^{-}$
WFG6	100	2	11	100000	8.4645×10^{-2} '='	8.5944×10^{-2} '='	9.0430×10^{-2} '−'	8.1863×10^{-2} '='	$8.2999 \times 10^{-}$
WFG7	100	2	11	100000	1.3391×10^{-2} '+'	1.3672×10^{-2} '='	3.2030×10^{-2} '−'	1.6980×10^{-2} '−'	$1.2234 \times 10^{-}$
WFG8	100	2	11	100000	1.0741×10^{-1} '='	1.0690×10^{-1} '='	1.1317×10^{-1} '−'	1.0893×10^{-1} '−'	$1.0541 \times 10^{-}$
WFG9	100	2	11	100000	1.6238×10^{-2} '='	1.5510×10^{-2} '='	3.2450×10^{-2} '−'	1.8524×10^{-2} '−'	$1.4531 \times 10^{-}$
MaF1	100	2	11	100000	3.7724×10^{-3} '+'	3.8007×10^{-3} '='	3.9789×10^{-3} '−'	4.6277×10^{-3} '−'	$3.5707 \times 10^{-}$
MaF2	100	2	11	100000	2.1457×10^{-3} '+'	2.1783×10^{-3} '='	2.3809×10^{-3} '−'	2.7588×10^{-3} '−'	$2.0068 \times 10^{-}$
MaF3	100	2	11	100000	4.3975×10^{-3} '='	4.3869×10^{-3} '='	7.7584×10^{-3} '−'	4.9079×10^{-3} '−'	$6.4275 \times 10^{-}$
MaF4	100	2	11	100000	1.2679×10^{-2} '='	1.2608×10^{-2} '='	4.0880×10^{-2} '−'	1.5525×10^{-2} '−'	$3.5083 \times 10^{-}$
MaF5	100	2	11	100000	1.2946×10^{-2} '='	1.2871×10^{-2} '='	3.2827×10^{-2} '−'	1.5951×10^{-2} '−'	$1.2231 \times 10^{-}$
MaF6	100	2	11	100000	4.0666×10^{-3} '+'	4.0761×10^{-3} '='	1.0025×10^{-2} '−'	4.9620×10^{-3} '−'	$3.9661 \times 10^{-}$
MaF7	100	2	21	100000	4.7520×10^{-3} '='	4.7574×10^{-3} '='	5.1753×10^{-3} '−'	5.4252×10^{-3} '−'	$5.1169 \times 10^{-}$

The distributional behavior of IGD values is further illustrated in Fig. 2, which presents boxplots for all $M = 2$ instances. Visually, IVF/SPEA2 exhibits narrower interquartile ranges than SPEA2 on ZDT and DTLZ problems, suggesting more consistent convergence. On WFG instances, the distributions overlap substantially, reflecting the statistical ties reported in Table 3. Among MaF problems, MaF1 and MaF2 show the clearest median separation in favor of IVF/SPEA2.

For tri-objective problems, IVF/SPEA2 continues to show improvements in a majority of instances while also exposing problem-dependent limitations, as reflected in the observed wins and losses relative to SPEA2.

Table 4 Comparison of IGD (median) between IVF/SPEA2 and baselines for $M = 3$ (DTLZ1–DTLZ7, WFG1–WFG9, and MaF1–MaF7). Symbols $+$, $-$, and $=$ denote statistically significant superiority, inferiority, or equivalence relative to SPEA2 under the Wilcoxon rank-sum test ($p < 0.05$).

Problem	N	M	D	FE	IVF/SPEA2	MFO-SPEA2	SPEA2+SDE	NSGA-II	
DTLZ1	100	3	7	100000	$2.0112 \times 10^{-2}\,'='$	$2.0184 \times 10^{-2}\,'-'$	$2.0574 \times 10^{-2}\,'-'$	$2.7057 \times 10^{-2}\,'-'$	2.0570
DTLZ2	100	3	12	100000	$5.3965 \times 10^{-2}\,'+'$	$5.4223 \times 10^{-2}\,'-'$	$7.6732 \times 10^{-2}\,'='$	$6.9228 \times 10^{-2}\,'-'$	5.4464
DTLZ3	100	3	12	100000	$5.3168 \times 10^{-2}\,'+'$	$5.3458 \times 10^{-2}\,'-'$	$7.6058 \times 10^{-2}\,'='$	$6.9016 \times 10^{-2}\,'-'$	5.4699
DTLZ4	100	3	12	100000	$3.8805 \times 10^{-1}\,'='$	$1.5063 \times 10^{-1}\,'+'$	$1.9838 \times 10^{-1}\,'='$	$6.7606 \times 10^{-2}\,'+'$	1.6540
DTLZ5	100	3	12	100000	$4.3156 \times 10^{-3}\,'+'$	$1.2417 \times 10^{-2}\,'-'$	$1.0011 \times 10^{-2}\,'='$	$5.2873 \times 10^{-3}\,'-'$	3.3906
DTLZ6	100	3	12	100000	$4.0831 \times 10^{-3}\,'='$	$4.0876 \times 10^{-3}\,'-'$	$9.6308 \times 10^{-3}\,'='$	$5.8744 \times 10^{-3}\,'-'$	1.8730
DTLZ7	100	3	22	100000	$6.9223 \times 10^{-2}\,'+'$	$6.0497 \times 10^{-2}\,'='$	$8.0552 \times 10^{-2}\,'='$	$8.2232 \times 10^{-2}\,'-'$	8.1043
WFG1	100	3	12	100000	$1.5019 \times 10^{-1}\,'+'$	$1.5289 \times 10^{-1}\,'-'$	$1.7729 \times 10^{-1}\,'='$	$2.2384 \times 10^{-1}\,'-'$	1.4904
WFG2	100	3	12	100000	1.7885×10^{-1}	$1.6847 \times 10^{-1}\,'-'$	$2.0638 \times 10^{-1}\,'+'$	$2.2266 \times 10^{-1}\,'-'$	1.6469
WFG3	100	3	12	100000	$1.0839 \times 10^{-2}\,'+'$	$1.0472 \times 10^{-2}\,'+'$	$1.6581 \times 10^{-2}\,'-'$	$1.5475 \times 10^{-2}\,'+'$	9.3581
WFG4	100	3	12	100000	$2.1409 \times 10^{-1}\,'+'$	$2.1581 \times 10^{-1}\,'-'$	$3.2292 \times 10^{-1}\,'='$	$2.6985 \times 10^{-1}\,'-'$	2.0092
WFG5	100	3	12	100000	$2.2190 \times 10^{-1}\,'+'$	$2.2265 \times 10^{-1}\,'-'$	$3.3187 \times 10^{-1}\,'='$	$2.7858 \times 10^{-1}\,'-'$	2.2989
WFG6	100	3	12	100000	$2.4118 \times 10^{-1}\,'='$	$2.4237 \times 10^{-1}\,'-'$	$3.4176 \times 10^{-1}\,'='$	$3.0460 \times 10^{-1}\,'-'$	2.3889
WFG7	100	3	12	100000	$2.1575 \times 10^{-1}\,'+'$	$2.1802 \times 10^{-1}\,'-'$	$3.3027 \times 10^{-1}\,'='$	$2.7874 \times 10^{-1}\,'-'$	2.2109
WFG8	100	3	12	100000	$2.9318 \times 10^{-1}\,'+'$	$2.9803 \times 10^{-1}\,'-'$	$3.5941 \times 10^{-1}\,'-'$	$3.7429 \times 10^{-1}\,'-'$	2.7899
WFG9	100	3	12	100000	2.1691×10^{-1}	$2.9198 \times 10^{-1}\,'-'$	$1.3367 \times 10^{-1}\,'='$	$2.7314 \times 10^{-1}\,'-'$	2.2159
MaF1	100	3	12	100000	$4.1484 \times 10^{-2}\,'+'$	$4.1895 \times 10^{-2}\,'-'$	$4.2174 \times 10^{-2}\,'='$	$5.7330 \times 10^{-2}\,'-'$	6.0677
MaF2	100	3	12	100000	$3.2614 \times 10^{-2}\,'='$	$3.0998 \times 10^{-2}\,'+'$	$3.1149 \times 10^{-2}\,'+'$	$4.5892 \times 10^{-2}\,'-'$	3.6002
MaF3	100	3	12	100000	$3.2863 \times 10^{-2}\,'+'$	$3.3571 \times 10^{-2}\,'-'$	$5.8761 \times 10^{-2}\,'='$	$1.0803 \times 10^{-1}\,'-'$	4.7182
MaF4	100	3	12	100000	$2.4014 \times 10^{-1}\,'+'$	$3.6409 \times 10^{-1}\,'-'$	$3.6408 \times 10^{-1}\,'='$	$8.6124 \times 10^{-1}\,'-'$	4.0161
MaF5	100	3	12	100000	1.5179×10^{0}	$4.8561 \times 10^{-1}\,'-'$	$8.3612 \times 10^{-1}\,'='$	$7.5943 \times 10^{-1}\,'-'$	8.5360
MaF6	100	3	12	100000	$4.0636 \times 10^{-3}\,'+'$	$4.0809 \times 10^{-3}\,'-'$	$9.7715 \times 10^{-3}\,'='$	$5.2715 \times 10^{-3}\,'-'$	1.5222
MaF7	100	3	22	100000	$7.4311 \times 10^{-2}\,'='$	$6.5112 \times 10^{-2}\,'-'$	$5.9557 \times 10^{-2}\,'='$	$7.7718 \times 10^{-2}\,'-'$	9.6487

Fig. 3 presents boxplots for the tri-objective setting. The visual patterns corroborate the tabular results: IVF/SPEA2 achieves visibly lower medians on DTLZ2, DTLZ3, and DTLZ5, while DTLZ4 presents high variability for all algorithms. Among MaF problems, MaF4 stands out with IVF/SPEA2 achieving a markedly lower median, whereas MaF5 reveals a case where IVF/SPEA2 underperforms, consistent with the loss reported in Table 4.

6 Discussion

This study evaluates IVF/SPEA2 as a memetic enhancement to SPEA2. The empirical results indicate that integrating the IVF mechanism can improve performance across several benchmark instances, but the magnitude and consistency of gains depend on problem characteristics.

6.1 Observed Performance Patterns

Across 79 benchmark instances, IVF/SPEA2 demonstrates improved or competitive performance in many cases. In bi-objective scenarios ($M = 2$), IVF/SPEA2 achieves statistical superiority or equivalence in all 28 instances, with 16 significant improvements (57.1%) and no statistically significant degradations. For tri-objective problems ($M = 3$), IVF/SPEA2 obtains 14 wins (60.9%) against 3 losses (13.0%). These outcomes support the hypothesis that controlled intensification can complement the host algorithm under a fixed evaluation budget, while also highlighting problem-dependent behavior.

6.2 Comparative Behavior vs. Baselines

When compared to established algorithms (NSGA-II, NSGA-III, MOEA/D, MFO-SPEA2, SPEA2+SDE), IVF/SPEA2 remains competitive. Notably, the method shows advantages over NSGA-III in tri-objective scenarios where NSGA-III's performance declines. However, the comparative advantages remain modest and depend on the benchmark structure, reinforcing the understanding that no single algorithm dominates across all problem types.

6.3 Effect Size and Practical Significance

Beyond statistical significance, Vargha–Delaney A_{12} effect size analysis indicates practical differences in certain categories. Fig. 4 summarizes the A_{12} values across all benchmark instances, grouped by suite and objective count. Stronger effects are observed on problems featuring regular Pareto front geometries (e.g., MaF1, MaF3, MaF6 with $A_{12} > 0.71$), as visually confirmed by the darker cells in Fig. 4. These favorable outcomes, however, are limited to specific structures and should not be generalized to all scenarios.

6.4 Mechanistic Interpretation

The observed behavior suggests that IVF's targeted crossover operations can mitigate evolutionary stagnation in certain landscapes. With $c = 10\%$ and $r = 10\%$, the intervention strategy maintains evolutionary pressure while potentially preventing stagnation on specific problem types. The approach shows particular effectiveness on problems with linear, convex, or degenerate Pareto fronts (e.g., DTLZ1, DTLZ2, DTLZ5, DTLZ6).

Conversely, limitations emerge on highly deceptive landscapes (e.g., MaF2, WFG2), where combined elitist mechanisms may contribute to premature convergence. This reinforces that IVF/SPEA2 is not universally beneficial, and that its suitability depends on structural properties of the optimization landscape.

Overall, the results provide evidence that targeted memetic enhancements can improve SPEA2 in specific contexts, while emphasizing the importance of matching algorithmic mechanisms to problem characteristics and practical constraints.

7 Conclusion

This work investigates IVF/SPEA2, a hybrid algorithm that integrates the In Vitro Fertilization (IVF) mechanism into SPEA2 as a memetic enhancement. The empirical evaluation indicates that IVF/SPEA2 can improve SPEA2 performance across multiple benchmark instances, but the benefits are problem-dependent rather than universal.

Across bi-objective instances ($M = 2$), IVF/SPEA2 achieves statistical superiority or equivalence in all evaluated cases, with 16 significant improvements and no statistically significant degradations. For tri-objective problems ($M = 3$), the method obtains more wins than losses, indicating that the controlled coupling strategy can be effective under fixed evaluation budgets. Comparative analyses against established algorithms further position IVF/SPEA2 as competitive, particularly in tri-objective scenarios where some baselines exhibit performance decline.

The findings also delineate the scope of the approach: IVF/SPEA2 tends to perform better on problems with regular Pareto front geometries and may struggle on highly deceptive landscapes where elitist dynamics can lead to premature convergence. Accordingly, the method should be viewed as a modest enhancement to SPEA2 that is advantageous in specific problem classes.

Future research directions include adaptive control of c and r based on diversity or convergence signals, validation on real-world optimization applications, and the investigation of alternative memetic components or hybrid combinations to broaden robustness across diverse landscapes.

Supplementary information.

Acknowledgements.

Declarations

- **Funding:** Not applicable.
- **Competing interests:** The authors declare no competing interests.
- **Ethics approval and consent to participate:** Not applicable.
- **Consent for publication:** Not applicable.
- **Data availability:** The data that support the findings of this study are available from the corresponding author upon reasonable request.
- **Materials availability:** Not applicable.
- **Code availability:** The code that supports the findings of this study is available from the corresponding author upon reasonable request.
- **Author contributions:** Not applicable.

Appendix A Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.

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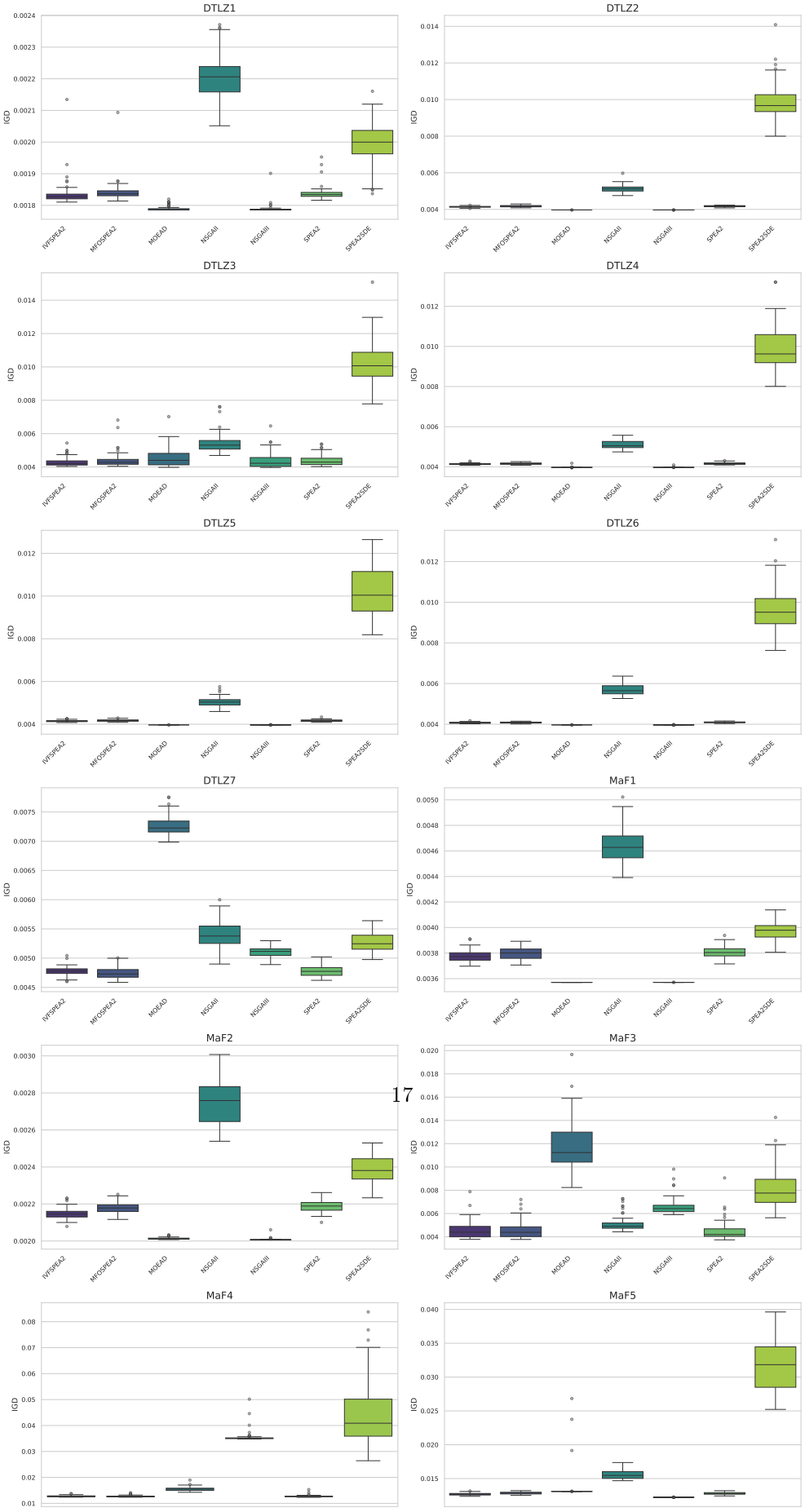
Distribuição IGD por Problema (M2)
(Execuções com IGD > 0.1 removidas)



Distribuição IGD por Problema (M3)
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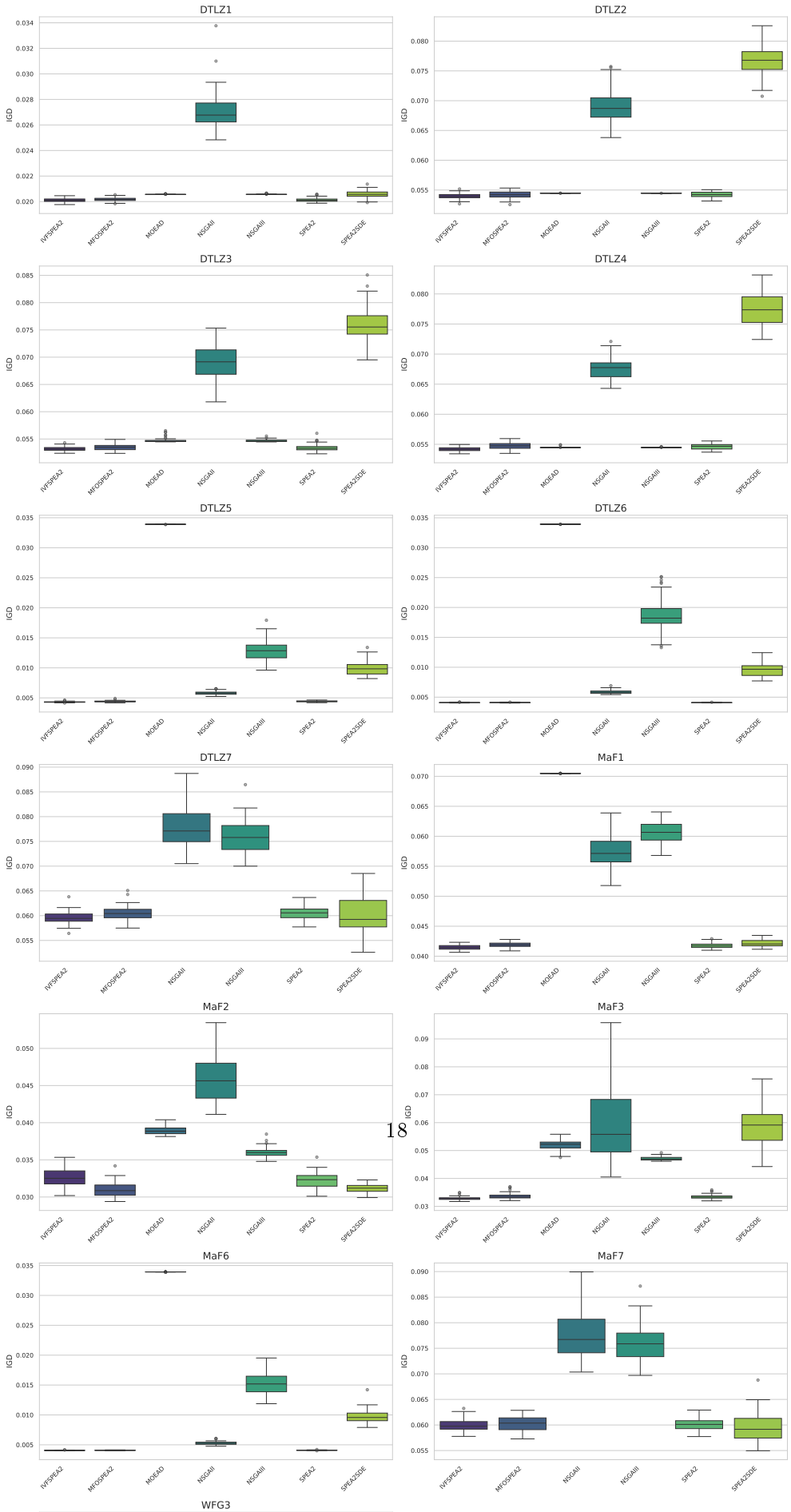

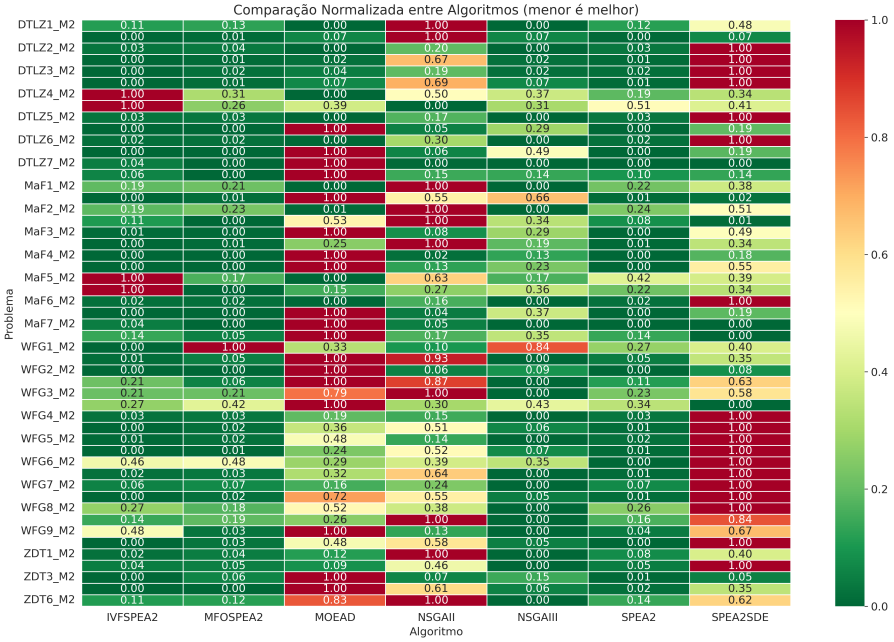

Fig. 4 Vargha–Delaney A_{12} effect size comparing IVF/SPEA2 against SPEA2 across benchmark instances, grouped by benchmark suite and objective count.